Regulatory Capture via Price Aggregator

Gabriel Cañedo Riedel*

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Abstract

This paper develops a theory of regulatory capture in which firms benefit from industry-wide regulation that raises their own costs. Under monopolistic competition with a broad class of demand systems, a firm's profit depends on its cost relative to an industry price aggregator—not on its cost in levels. Regulation that raises all costs can therefore improve the competitive position of the most productive firms if it shifts the price aggregator by more than it raises their costs. I classify cost-increasing regulations into three types—proportional, additive, and shape-changing—and show that only shape-changing regulations generate this effect. Proportional cost increases are neutral; additive cost increases harm the most productive firms.

I apply the theory to a well-documented puzzle in the minimum wage literature: after a minimum wage increase, firms already paying above the new floor expand, gain market share, and appear to become more productive. Existing explanations rely on labor market mechanisms—monopsony, search frictions, managerial slack—yet the firms that benefit are precisely those least affected through the labor market. The theory provides a product market explanation: because the minimum wage has a vanishing direct impact on high-productivity firms but raises costs for labor-intensive competitors, it operates as a shape-changing regulation. The price aggregator rises, relative prices shift, and demand reallocates toward the most productive firms within the industry. The resulting gains in market share and revenue are measured as productivity increases.

1 Introduction

1.1 Motivation

Why do firms sometimes advocate for regulation that raises their own costs? Large retailers have supported minimum-wage increases that raise labor costs across their sector. Technology firms have backed clean-energy mandates that raise the cost of computing for everyone. Incumbents in carbon-intensive industries have publicly committed to lobby for ambitious climate targets. In each case, the firm bears part of the cost it is advocating for.

This paper proposes a mechanism. Under a broad class of demand systems, the object that governs a firm's profit is not its marginal cost in levels, but its marginal cost relative to an industry-wide price aggregator. Since prices in monopolistic competition are proportional to marginal costs, this relative marginal cost maps directly to a relative price, which in turn determines market share and profit. A regulation that raises all marginal costs therefore need not harm all firms: if the price aggregator rises

*gabrielcr@ucsd.edu; Department of Economics at University of California, San Diego.

by more than a given firm's cost, that firm's relative price falls, its market share grows, and its profit increases—even though its own cost has risen.

A concrete instance makes the mechanism transparent. A large empirical literature documents that after minimum wage increases, firms already paying above the new floor expand, gain market share, and appear to become more productive. The pattern is robust across countries and settings. Yet existing explanations—monopsony power, search frictions, managerial slack—are fundamentally labor market mechanisms, and the firms that benefit are precisely those least constrained by the labor market. The theory developed here offers a product market alternative: the minimum wage has a vanishing direct impact on high-productivity firms but raises costs substantially for their labor-intensive competitors. It therefore operates as a shape-changing regulation. The price aggregator rises, relative prices shift within the industry, and demand reallocates toward the most productive firms. The resulting gains in revenue and market share are included in standard productivity measures, so they appear as productivity increases.

1.2 The price aggregator under HSA demand

Consider monopolistic competition with a continuum of firms $\omega \in (0, 1)$, each producing a differentiated variety with constant marginal cost $\psi_\omega > 0$. Demand is homothetic single-aggregator (HSA), as in Matsuyama and Ushchev (2020): demand for variety ω takes the form

$$x_\omega = \frac{Y}{p_\omega} s_\omega \left(\frac{p_\omega}{A(\mathbf{p})} \right), \tag{1}$$

where Y is aggregate expenditure, $s_\omega(\cdot)$ is a share function, and $A(\cdot)$ is a homothetic price aggregator. I assume that the share function is decreasing in relative price, $s'_\omega(z) < 0$, so that a firm that raises its price loses expenditure share. This class nests CES, translog, and other commonly used demand systems.

The key implication of HSA demand for firms is that maximizing over price p_ω is equivalent to maximizing over relative price $z_\omega \equiv p_\omega / A(\mathbf{p})$. Profit can be written as

$$\pi_\omega = (p_\omega - \psi_\omega) x_\omega \left(\frac{p_\omega}{A(\mathbf{p})} \right) = \left(1 - \frac{\psi_\omega / A(\mathbf{p})}{z_\omega} \right) Y s_\omega \left(z_\omega \right) \tag{2}$$

so equilibrium profit is strictly decreasing in the firm's **relative marginal cost** $\psi_\omega / A(\mathbf{p})$. Since equilibrium prices are $p^*_\omega = \mu_\omega \psi_\omega$, the price aggregator is $A(\boldsymbol{\mu\psi})$, and the relative marginal cost is the sufficient statistic for a firm's competitive position: it determines its relative price, its market share, and its profit.

A regulation changes the cost distribution from ψ to ψ^R. Firm ω benefits whenever

$$\frac{\psi^R_\omega}{A(\boldsymbol{\mu\psi}^R)} < \frac{\psi_\omega}{A(\boldsymbol{\mu\psi})}, \qquad \text{even if} \quad \psi^R_\omega > \psi_\omega. \tag{3}$$

Whether this can happen—and for which firms—depends on how the regulation changes the distribution of marginal costs.

1.3 Main results

To obtain sharp predictions, I classify regulations by how they deform the distribution of marginal costs. Let marginal costs be distributed according to a continuous parametric CDF, $\omega = G(\psi_\omega)$ with $G \in \mathcal{G}$.

Changes in the parameters of G fall into three types—scale, location, and shape—and each interacts differently with the price aggregator.

Scale (proportional). A scale change multiplies every marginal cost by the same factor:

$$\psi_\omega^R = r\,\psi_\omega \qquad \text{for all } \omega \in (0,1),\ r > 1.$$

These are isomorphic to ad valorem taxes or uniform input price increases. By homogeneity of the price aggregator, $A(r\boldsymbol{\mu\psi}) = rA(\boldsymbol{\mu\psi})$, so relative marginal costs are unchanged. **Scale regulation is neutral.**

Location (additive). A location change adds the same constant to every marginal cost:

$$\psi_\omega^R = \psi_\omega + a \qquad \text{for all } \omega \in (0,1),\ a > 0.$$

These are isomorphic to per-unit fees or flat compliance costs. An additive increase is proportionally larger for low-cost firms. **Additive regulation harms the most productive firms.**

Shape. A shape change deforms the distribution without changing its support. Unlike scale or location, it cannot be written as a uniform multiplication or addition. The defining feature is that a shape change can raise marginal costs for all firms while having a **vanishing marginal burden** on the most productive.

The minimum wage is a natural example. A minimum wage increase raises labor costs for every firm, but the most productive firms already pay well above the minimum wage. For them, the direct cost impact is negligible. For their less productive competitors, whose wages cluster near the floor, the impact is substantial.

Because the burden vanishes at the left tail, the price aggregator $A(\boldsymbol{\mu\psi}^R)$ rises while the most productive firms' costs barely change, leading to a decline in their relative marginal cost:

$$\frac{\psi_\omega^R}{A(\boldsymbol{\mu\psi}^R)} < \frac{\psi_\omega}{A(\boldsymbol{\mu\psi})}.$$

Shape-changing regulation benefits the most productive firms.

Together: proportional regulation is neutral, additive regulation harms the most productive, and capture from the top arises only from shape-changing regulation.

1.4 Illustration and application

I provide a closed-form illustration under CES demand with Pareto-distributed marginal costs. A marginal decrease in the Pareto shape parameter raises costs for all firms, increases the CES price index, and yields an explicit cutoff ε characterizing the set of firms that benefit. The cutoff depends only on the shape parameter κ and the elasticity of substitution σ, making the comparative statics transparent: a shape change can be negligible at the left tail while substantially raising the price index through the rest of the distribution.

The minimum wage application puts the theory to work. A large empirical literature documents that after minimum wage increases, high-productivity firms expand and appear to become more productive—a pattern typically attributed to labor market mechanisms. The model offers a product market alternative: the minimum wage operates as a shape-changing regulation, and the resulting shift in the price aggregator reallocates demand toward the most productive firms within the industry. The gains in revenue and market share are included in standard productivity measures, so they are recorded

as productivity increases. I discuss the testable predictions that distinguish this channel from existing explanations.

1.5 Relation to the literature

The paper connects three literatures.

Regulation as a traded good. The canonical political economy of regulation—Stigler (1971), Peltzman (1976), Becker (1983), Laffont and Tirole (1993), Grossman and Helpman (1994)—models the regulator as an agent and studies how organized interests compete to influence policy. The focus is on the supply side of regulation: what the regulator maximizes, how interest groups lobby, and what equilibrium policy emerges. Incentives within an industry are typically aligned; the conflict is across industries, not within them. This paper shifts the focus to the demand side: even taking the regulation as given, a uniform instrument creates winners and losers within an industry, determined by relative marginal cost $\psi_\omega/A(\boldsymbol{\mu\psi})$, not by marginal cost alone.

Raising rivals' costs. The RRC tradition—Salop and Scheffman (1983, 1987), building on the empirical observation of Marvel (1977)—shows how a firm can profit by raising its competitors' costs through regulation, standards, or input contracts. The canonical setting is finite-firm oligopoly with homogeneous goods, and the asymmetric incidence is baked into the regulation: exemptions, grandfathering, or targeted standards that hit rivals by design. Rents arise through exit or reduced entry—the departure of rivals raises the single market price, benefiting survivors. The mechanism in this paper differs in three ways: it operates under monopolistic competition with differentiated goods, requires no strategic interaction, and does not need the regulation itself to be asymmetric—the heterogeneous effective burden arises endogenously from differences in firm productivity under a uniform rule.

Firm size and capture. A growing empirical literature documents that large or politically connected firms gain disproportionately in regulated environments (Crane and Koch, 2025a; Fan and Zhou, 2023b; Huneeus and Kim, 2024). These papers typically model the benefits as idiosyncratic shocks to connected firms, leaving the underlying mechanism through which regulation generates rents largely unspecified. The present paper provides such a mechanism. On the theory side, Macedoni and Weinberger (2024) show that large firms can favor regulation that raises fixed costs. Their CES-Pareto framework implies that marginal-cost regulation is neutral—a direct application of Theorem 2. The present paper shows that marginal-cost regulation can redistribute when it changes the shape of the cost distribution.

Minimum wage and firm outcomes. The application connects to a large empirical literature documenting that minimum wage increases reallocate activity toward more productive firms (Rao and Risch, 2026; Dustmann et al., 2022; Engbom and Moser, 2022; Haelbig et al., 2023; Draca et al., 2011). Existing explanations rely on labor market mechanisms: monopsony (Berger et al., 2025; Haanwinckel, 2025), search frictions (Drechsel-Grau, 2023), or managerial slack (Hau et al., 2020; Mayneris et al., 2018). The few papers that study product market channels do so in frameworks with homogeneous goods, where relative prices between firms cannot change by construction (Rao and Risch, 2026; Williamson, 1968; Haucap et al., 2001). This paper provides the first product market mechanism in which differentiated goods and a heterogeneous cost shock generate demand reallocation within the industry.

1.6 Outline

Section 2 sets up the environment: monopolistic competition with HSA demand and the key role of relative marginal cost. Section 3 defines regulation as a change in the parametric family governing marginal costs and introduces the scale, location, and shape taxonomy. Section 4 states and proves the three main theorems. Section 5 provides a closed-form illustration under CES demand with Pareto costs. Section 6 applies the theory to the minimum wage. Section 7 reviews the related literature. Section 8 concludes.

2 Environment

Consider a static, closed economy with a unit continuum of differentiated varieties $\omega \in (0, 1)$.

Demand. Consumer preferences are homothetic single-aggregator (HSA), as in Matsuyama and Ushchev (2020). Demand for variety ω is

$$x_\omega = \frac{Y}{p_\omega} s_\omega\left(\frac{p_\omega}{A(\mathbf{p})}\right), \tag{4}$$

where $Y > 0$ is aggregate expenditure, $s_\omega(\cdot)$ is the share function, and $A(\mathbf{p})$ is a homothetic price aggregator implicitly defined by the adding-up condition

$$\int_0^1 s_\omega\left(\frac{p_\omega}{A(\mathbf{p})}\right) d\omega = 1. \tag{5}$$

I assume throughout that the share function is decreasing in relative price: $s'_\omega(z) < 0$ for all $z > 0$. This ensures that a firm raising its price loses expenditure share, and that $A(\mathbf{p})$ is strictly increasing in every p_ω. Under CES, this condition is equivalent to gross substitutes ($\sigma > 1$).

Firms. Each variety is produced by a monopolistically competitive firm with constant marginal cost $\psi_\omega > 0$. Firm ω chooses its price to maximize

$$\pi_\omega = (p_\omega - \psi_\omega)\, x_\omega. \tag{6}$$

Let $z_\omega \equiv p_\omega / A(\mathbf{p})$ denote the relative price of variety ω. Substituting and rearranging:

$$\pi_\omega = \left(1 - \frac{\psi_\omega / A(\mathbf{p})}{z_\omega}\right) Y\, s_\omega(z_\omega). \tag{7}$$

Since $A(\mathbf{p})$ is pinned down by (5), choosing p_ω is equivalent to choosing z_ω, and profit depends on the firm's own cost only through the *relative marginal cost* $\psi_\omega / A(\mathbf{p})$.

Equilibrium. An equilibrium is a price vector $\mathbf{p}^* = \boldsymbol{\mu\psi}$ satisfying each firm's first-order condition, where $\mu_\omega > 1$ is the markup. By the envelope theorem, equilibrium profit is strictly decreasing in relative marginal cost:

$$\frac{\partial \pi^*_\omega}{\partial(\psi_\omega / A(\boldsymbol{\mu\psi}))} = -\frac{Y\, s_\omega(z^*_\omega)}{z^*_\omega} < 0. \tag{8}$$

Market share $s_\omega(z^*_\omega)$ is likewise strictly decreasing in $\psi_\omega/A(\boldsymbol{\mu\psi})$. The relative marginal cost is the sufficient statistic for a firm's competitive position: it determines relative price, market share, and profit.

3 Regulation

Let marginal costs be distributed according to a continuous parametric family $\mathcal{G}$ of CDFs on $[\psi_0, \infty)$, $\psi_0 > 0$, with $\omega = G(\psi_\omega)$ for $G \in \mathcal{G}$. I assume that the support of G does not depend on the full parameter vector, so that changes in scale, location, and shape parameters can be cleanly separated.[1]

Definition 1. *A **regulation** R is a change in marginal costs from $\psi \sim G$ to $\psi^R \sim G^R$, with $G, G^R \in \mathcal{G}$.*

I classify cost-increasing regulations by which parameter of G moves.

Definition 2. *A regulation R is **cost-increasing in scale** (R-cic) if $\psi^R_\omega = r\,\psi_\omega$ for all $\omega \in (0,1)$ and some $r > 1$.*

Scale regulations are isomorphic to ad valorem taxes or proportional input price increases.

Definition 3. *A regulation R is **cost-increasing in location** (R-cil) if $\psi^R_\omega = \psi_\omega + a$ for all $\omega \in (0,1)$ and some $a > 0$.*

Location regulations are isomorphic to per-unit fees or flat compliance costs. The proportional cost increase $(\psi_\omega + a)/\psi_\omega = 1 + a/\psi_\omega$ is larger for low-cost firms, so location shifts compress the cost distribution from below.

Definition 4. *A regulation R is **cost-increasing in shape** (R-cis) if $\psi^R_\omega > \psi_\omega$ for all $\omega \in (0,1)$ through a change in the shape parameter κ.*

Shape regulations deform the distribution without changing its support. By continuity of $\mathcal{G}$ and support invariance, a shape change produces proportional cost increases that converge to one at the left tail:

$$\lim_{\omega \downarrow 0} \frac{\psi^R_\omega}{\psi_\omega} = 1. \tag{9}$$

That is, the marginal burden of a shape-changing regulation vanishes for the most productive firms.

4 Main Results

The three theorems below characterize how each type of cost-increasing regulation affects equilibrium outcomes. The central object in each case is the change in relative marginal cost $\psi_\omega/A(\boldsymbol{\mu\psi})$ following regulation.

Theorem 1 (Shape increases benefit the most productive firms)**.** *Let R be R-cis. Then there exists $\varepsilon > 0$ such that for all $\omega \in (0, \varepsilon)$:*

$$\frac{\psi^R_\omega}{A(\boldsymbol{\mu\psi}^R)} < \frac{\psi_\omega}{A(\boldsymbol{\mu\psi})}.$$

Consequently, firms in $(0, \varepsilon)$ attain strictly higher equilibrium profits and market shares following R.

[1] This condition is satisfied by the vast majority of parametric families used in economics, including: Normal, Log-normal, Gamma, Weibull, Beta, Pareto, Log-logistic, Burr, Fréchet, Student's t, Chi-squared, F, Cauchy, Logistic, Laplace, Gumbel, Inverse Gaussian, and Exponential. The notable exceptions—distributions whose support depends on a shape or scale parameter, such as the Uniform—are rare in practice.

Proof. Since $\psi^R_\omega > \psi_\omega$ for all ω and $A(\cdot)$ is strictly increasing in its arguments, the equilibrium price index rises:

$$A(\boldsymbol{\mu\psi}^R) > A(\boldsymbol{\mu\psi}).$$

By (9), $\psi^R_\omega/\psi_\omega \to 1$ as $\omega \downarrow 0$. Since the ratio $A(\boldsymbol{\mu\psi}^R)/A(\boldsymbol{\mu\psi}) > 1$ is a fixed constant, there exists $\varepsilon > 0$ such that for all $\omega \in (0, \varepsilon)$:

$$\frac{\psi^R_\omega}{\psi_\omega} < \frac{A(\boldsymbol{\mu\psi}^R)}{A(\boldsymbol{\mu\psi})},$$

which rearranges to $\psi^R_\omega/A(\boldsymbol{\mu\psi}^R) < \psi_\omega/A(\boldsymbol{\mu\psi})$. The conclusion on profits and shares follows from (8). □

Remark 1 (Minimal conditions). The parametric structure is sufficient but not necessary. The result holds under two primitive conditions alone: (i) strict aggregator ordering, $A(\boldsymbol{\mu\psi}^R) > A(\boldsymbol{\mu\psi})$; and (ii) left-tail convergence, $\lim_{\omega\downarrow 0} \psi^R_\omega/\psi_\omega = 1$. No shared parametric family, no stochastic dominance between ψ and ψ^R, and no specific functional form for $A(\cdot)$ are required beyond monotonicity.

Corollary 1 (Exit amplifies)**.** *Suppose R is R-cis and choke prices are sufficiently binding that a positive measure of firms exits: $|\Omega^R| < |\Omega|$. Then surviving firms in $(0, \varepsilon) \cap \Omega^R$ reap strictly larger gains than in the no-exit case.*

Proof. From Theorem 1, in the absence of exit, $A(\boldsymbol{\mu\psi}^R) > A(\boldsymbol{\mu\psi})$. With exit, the post-regulation price aggregator $A^R(\cdot)$ is implicitly defined over the smaller active set Ω^R:

$$\int_{\Omega^R} s_\omega\left(\frac{\mu_\omega\psi_\omega}{A^R(\boldsymbol{\mu\psi}^R)}\right) d\omega = 1.$$

With fewer active firms and unchanged share functions ($s'_\omega < 0$), the adding-up condition requires $A^R(\boldsymbol{\mu\psi}^R) > A(\boldsymbol{\mu\psi}^R)$. Hence for all surviving $\omega \in (0, \varepsilon) \cap \Omega^R$:

$$\frac{\psi^R_\omega}{A^R(\boldsymbol{\mu\psi}^R)} < \frac{\psi^R_\omega}{A(\boldsymbol{\mu\psi}^R)} < \frac{\psi_\omega}{A(\boldsymbol{\mu\psi})},$$

which strictly improves on the no-exit case. □

Exit is therefore not a prerequisite for capture: the mechanism operates through the price aggregator alone. Exit amplifies the gains when it occurs because it forces the price index to adjust further to re-normalize expenditure shares over the smaller set of active firms.

Theorem 2 (Scale increases are neutral)**.** *Let R be R-cic. Then $\psi^R_\omega/A(\boldsymbol{\mu\psi}^R) = \psi_\omega/A(\boldsymbol{\mu\psi})$ for all $\omega \in (0, 1)$.*

Proof. A scale change sets $\psi^R_\omega = r\psi_\omega$ for all ω. By homogeneity of degree 1 of the price aggregator:

$$A(r\boldsymbol{\mu\psi}) = r\,A(\boldsymbol{\mu\psi}).$$

Therefore $\psi^R_\omega/A(\boldsymbol{\mu\psi}^R) = r\psi_\omega/(rA(\boldsymbol{\mu\psi})) = \psi_\omega/A(\boldsymbol{\mu\psi})$. □

Theorem 2 pins down a clean boundary: if a regulation is proportional — isomorphic to a uniform ad valorem tax — the price aggregator scales by exactly the same factor, and all relative positions are unchanged. There is no redistribution through the aggregator.

Theorem 3 (Location shifts harm the most productive firms)**.** *Let R be R-cil with increment $a > 0$. Let $\psi_\omega = G^{-1}(\omega)$ be continuous and strictly increasing, with $\psi_\omega \geq \psi_0 > 0$ and $\psi_\omega \to \infty$ as $\omega \uparrow 1$. Then there exists a unique $\varepsilon \in (0,1)$ such that firms in $(0,\varepsilon)$ are strictly harmed: $\psi^R_\omega / A(\boldsymbol{\mu}\boldsymbol{\psi}^R) > \psi_\omega / A(\boldsymbol{\mu}\boldsymbol{\psi})$.*

Proof. Define $\rho(a) \equiv A(\boldsymbol{\mu}(\boldsymbol{\psi} + a\mathbf{1}))/A(\boldsymbol{\mu}\boldsymbol{\psi})$ for $a > 0$.

Step 1 (bounds on $\rho(a)$). Monotonicity of A gives $\rho(a) > 1$. Since $\psi_\omega \geq \psi_0$, we have $\psi_\omega + a \leq (1 + a/\psi_0)\psi_\omega$ for all ω; by monotonicity and homogeneity of A:

$$\rho(a) \leq 1 + \frac{a}{\psi_0}.$$

Step 2 (IVT). Define $g(\omega) = (\psi_\omega + a)/\psi_\omega = 1 + a/\psi_\omega$. Then g is continuous and strictly decreasing with $\lim_{\omega \downarrow 0} g(\omega) = 1 + a/\psi_0$ and $\lim_{\omega \uparrow 1} g(\omega) = 1$. Since $\rho(a) \in (1,\, 1 + a/\psi_0]$, the intermediate value theorem guarantees a unique $\varepsilon \in (0,1)$ such that $g(\varepsilon) = \rho(a)$, i.e.,

$$\frac{A(\boldsymbol{\mu}(\boldsymbol{\psi} + a\mathbf{1}))}{A(\boldsymbol{\mu}\boldsymbol{\psi})} = \frac{\psi_\varepsilon + a}{\psi_\varepsilon}.$$

Step 3. For $\omega < \varepsilon$: $g(\omega) > g(\varepsilon) = \rho(a)$, which gives $(\psi_\omega + a)/\psi_\omega > A(\boldsymbol{\mu}\boldsymbol{\psi}^R)/A(\boldsymbol{\mu}\boldsymbol{\psi})$, hence $\psi^R_\omega / A(\boldsymbol{\mu}\boldsymbol{\psi}^R) > \psi_\omega / A(\boldsymbol{\mu}\boldsymbol{\psi})$. □

Together, Theorems 1–3 establish the complete boundary map: scale is neutral, location harms the most productive firms, and shape is the only channel through which pro-top-firm capture can arise.

Remark 2 (Cost-decreasing regulation). All three results have symmetric counterparts for cost-decreasing regulation. In particular, a shape *decrease* — a deregulation that lowers marginal costs with vanishing relief at the left tail — harms the most productive firms and benefits the least productive. Once a shape-changing regulation is in place, the firms that gained from it have an incentive to oppose its removal.

5 Illustration: CES Demand with Pareto Costs

This section provides a closed-form illustration under CES demand and Pareto marginal costs.

Setup. Let demand be CES with elasticity $\sigma > 1$, so the equilibrium price aggregator is

$$A(\boldsymbol{\mu}\boldsymbol{\psi}) = \left[\int_0^1 \left(\frac{\sigma}{\sigma - 1} \psi_\gamma \right)^{1-\sigma} d\gamma \right]^{\frac{1}{1-\sigma}}.$$

Assume marginal costs follow a Pareto distribution, $\psi \sim \mathrm{Pa}(\psi_0, \kappa)$, so that $\psi_\omega = \psi_0(1-\omega)^{-1/\kappa}$ for $\omega \in (0,1)$. The shape parameter $\kappa > \sigma - 1$ governs the fatness of the right tail; lower κ means higher dispersion.

Relative marginal cost. Substituting the Pareto quantile function into the CES aggregator gives

$$\frac{\psi_\omega}{\mu A(\boldsymbol{\psi})} = \frac{(1-\omega)^{-1/\kappa}}{\mu \left[\dfrac{\kappa}{\kappa + \sigma - 1} \right]^{\frac{1}{1-\sigma}}}. \tag{10}$$

The numerator varies by firm through $(1-\omega)^{-1/\kappa}$; the denominator is an industry-level term that depends on the shape of the cost distribution.

A shape-type regulation. Consider a regulation R that marginally decreases κ. Under Pareto costs, lower κ means higher dispersion: all firms' marginal costs rise, but the distance between firms widens. This is R-cis: costs increase for everyone, yet the proportional increase vanishes at the left tail.

What does a marginal decrease in κ look like in practice? Consider a new regulatory compliance requirement — an additional reporting obligation, a procedural filing, a mandatory audit. The regulation applies uniformly to all firms. But the most productive firms typically already maintain dedicated compliance departments, legal counsel, and administrative infrastructure; for them, the incremental cost of one more filing is negligible. Smaller, less productive firms must hire outside counsel or divert managerial attention, incurring a cost that is substantial relative to their existing operations. The burden is formally identical across firms, but its effective incidence vanishes at the top of the productivity distribution — exactly the signature of a decrease in κ.

A firm ω benefits if and only if its relative marginal cost falls, i.e., $\partial \ln(\psi_\omega / A(\boldsymbol{\psi}))/\partial\kappa > 0$. Differentiating (10) and setting equal to zero yields a closed-form cutoff:

$$\varepsilon = 1 - \exp\left(-\frac{\kappa}{\kappa+\sigma-1}\right). \tag{11}$$

All firms $\omega \in (0, \varepsilon)$ benefit; all firms $\omega \in (\varepsilon, 1)$ are harmed. The cutoff ε is strictly positive (consistent with Theorem 1) and strictly less than one.

Comparative statics. The cutoff ε is increasing in the CES elasticity σ: when goods are more substitutable, demand reallocation is stronger, and a larger fraction of firms benefits. The cutoff is decreasing in κ: fatter-tailed cost distributions generate larger asymmetries in the cost increase, shifting more of the burden to the right tail and enlarging the set of benefiting firms. As $\sigma \to \infty$, $\varepsilon \to 1 - 1/e \approx 0.63$; as $\kappa \to \infty$, $\varepsilon \to 0$.

6 Application: Minimum Wage

The capture mechanism applies directly to minimum wage policy. The standard view treats the minimum wage as a labor market instrument, but the framework above implies it operates, at least partly, through product market competition. This section develops the argument, explains why existing frameworks fall short, and derives testable predictions.

6.1 The puzzle

A robust empirical pattern across countries and settings is that minimum wage increases are followed by a reallocation of workers and output toward high-productivity firms. Workers flow from low- to high-productivity employers (e.g., Rao and Risch, 2026; Dustmann et al., 2022; Engbom and Moser, 2022; Haelbig et al., 2023). High-productivity firms expand employment, revenues, and measured productivity; low-productivity firms contract (e.g., Rao and Risch, 2026; Du and Wang, 2020; Mayneris et al., 2018; Draca et al., 2011; Luca and Luca, 2018).

The puzzle is not that some firms are hurt. It is that **firms already paying above the minimum wage** — firms for whom the regulation is mechanically inert — expand, gain market share, and appear more productive. Standard incidence logic offers no explanation: a firm unaffected by the wage floor should be unaffected, full stop.

6.2 Why labor market stories fall short

The literature proposes four explanations, each grounded in labor market mechanisms.

Monopsony and search frictions. The dominant explanation invokes either wage markdowns (Berger et al., 2025; Dustmann et al., 2022; Haanwinckel, 2025) or search congestion (Dustmann et al., 2022; Engbom and Moser, 2022; Drechsel-Grau, 2023): the minimum wage forces low-productivity firms to cut employment or vacancies, reducing congestion and enabling high-productivity firms to recruit more easily. Both stories require substantial labor market frictions. But minimum wage workers are in thick, low-skill labor markets with standardized, easily substitutable jobs — precisely the conditions under which frictions are smallest. Near-complete price pass-through (Allegretto and Reich, 2018; Harasztosi and Lindner, 2019) and price-to-wage ratios close to one (Aaronson and French, 2007) further undercut the monopsony premise.

Managerial slack and efficiency wages. A second strand (Hau et al., 2020; Mayneris et al., 2018; Riley and Rosazza Bondibene, 2017) argues that the wage shock forces organizational improvements or motivates workers through higher pay. Three difficulties arise. First, if efficiency gains were available, profit-maximizing firms would already implement them without a wage shock. Second, both mechanisms are one-sided: they predict gains only for firms directly affected by the floor, not for high-productivity firms already paying above it — yet those firms also expand (Draca et al., 2011; Haelbig et al., 2023). A firm whose wages were untouched by the policy should not experience more motivated workers or better management. Third, the minimum wage directly affects only low-wage workers, so firm-wide productivity gains would require implausibly large complementarities between low-skill and high-skill labor.

Common limitation. All four stories transmit the minimum wage through the labor market. Yet the most striking feature of the evidence is that firms already paying above the floor — firms whose labor market position was mechanically unaffected — also expand and gain. The product market mechanism explains this directly.

6.3 Minimum wage as a shape-type regulation

Under the framework of Section 2, the minimum wage operates as an R-cis regulation. All firms in the industry face higher labor costs, but the most productive firms already pay well above the minimum wage. For them, the direct cost impact is negligible. For their less productive competitors, whose wages cluster near the floor, the impact is substantial. Formally:

$$\lim_{\omega \downarrow 0} \frac{\psi_\omega^R}{\psi_\omega} = 1,$$

where the left tail indexes the most productive firms.

By Theorem 1, the industry-wide price aggregator rises, but the relative prices of high-productivity firms *fall*. Demand reallocates toward them. Their revenues and market shares expand even though they are mechanically unaffected by the floor.

The key implication concerns measurement. Standard productivity measures — labor productivity, revenue TFP, value added per worker — include profit as a component. Labor productivity, for instance, is (remuneration + profit) / employment; revenue TFP and value added per worker similarly reflect the firm's revenue position, not only its physical efficiency. When a firm gains market share and revenue because its relative price fell, its profit rises, and this shows up mechanically as a productivity increase. The observed gains at high-productivity firms are competitive position effects, not genuine efficiency improvements. Standard productivity decompositions cannot distinguish the two.

6.4 Why existing product market frameworks fall short

A small literature attributes minimum wage effects to product market mechanisms (Williamson, 1968; Haucap et al., 2001; Rao and Risch, 2026). These papers follow the Raising Rivals' Costs tradition and use Cournot competition with **homogeneous goods**. With homogeneous goods, all firms sell at a single market price; there is no relative price channel, and no individual firm can gain market share through a differential in price levels. Demand reallocation through relative prices is not expressible in this framework.

The one empirical paper that implicitly requires differentiated goods — Harasztosi and Lindner (2019) — imposes a proportional cost shock. That corresponds to an R-cic regulation, which by Theorem 2 is neutral with respect to relative prices. No redistribution through the aggregator can arise. The present paper resolves both limitations: differentiated goods with a shape-type cost shock.

6.5 Testable predictions

The product market mechanism generates sharp predictions that are distinct from all labor market stories discussed above. All are currently untested.[2]

1. **Relative prices.** After a minimum wage increase, low-productivity firms raise their prices more than high-productivity firms. The cross-firm distribution of price changes should be positively correlated with labor intensity.

2. **Within-industry market shares.** High-productivity firms gain market share within the industry. The key test is that this reallocation correlates with relative price changes, not with labor market variables.

3. **Employment follows output.** Employment at firms already paying above the minimum wage rises — not because of reduced labor market frictions, but because demand for their output expanded.

4. **Productivity is a competitive position effect.** Revenue-based TFP gains at high-productivity firms should be accounted for by their market share gain, with no residual improvement in physical productivity.

These predictions can be tested with employer-employee matched data combined with firm-level price or revenue information, using minimum wage variation across states or countries as identification.

[2] Restaurant-sector papers (Aaronson and French, 2007; Allegretto and Reich, 2018) are the only within-industry studies, but they assume homogeneous goods or putty-clay technology and draw no conclusions about relative prices or productivity redistribution across firms.

7 Literature Review

[This section is in progress.]

8 Conclusion

This paper proposes a new channel of regulatory capture. Under a broad class of demand systems, a firm's competitive position depends on its marginal cost relative to an industry-wide price aggregator — not on its marginal cost in levels. A regulation that raises all marginal costs can therefore reduce the relative costs of the most productive firms, if its effective burden vanishes at the left tail of the cost distribution. I classify cost-increasing regulations into three types and show that only shape-changing regulations produce this effect: proportional regulations are neutral, additive regulations harm the most productive firms.

The minimum wage application illustrates the practical reach of the mechanism. The most productive firms already pay well above the minimum wage, so a minimum wage increase operates as a shape-changing regulation: the direct cost impact is negligible for firms at the top but substantial for their less productive competitors. The price aggregator rises, relative prices shift, and demand reallocates toward the most productive firms within the industry. What appears in the data as a productivity gain at high-productivity firms is a competitive position effect — profit is a component of standard productivity measures.

The theory is currently positive: it characterizes when and which firms benefit from a given regulation. Two extensions follow naturally. First, the strategic side: under what conditions does a firm find it profitable to lobby for a shape-changing regulation, and how does the within-industry distribution of winners and losers shape the political equilibrium? Second, the quantitative side: the CES-Pareto example delivers a closed-form cutoff ε in (11) that can be calibrated, mapping directly into predictions about the magnitude and distribution of rents from minimum wage increases.

Appendix: Shape-changing regulations in practice

A regulation is shape-changing if it raises marginal costs for all firms but its burden becomes negligible for the most productive. This appendix documents that such instruments are widespread. Table 1 organizes six classes by the mechanism through which the burden vanishes at the left tail.

Table 1: Shape-changing regulations: mechanisms and examples

Category	**Why burden vanishes at the top**	**Examples**
Output-linked reporting	Large firms automate compliance and integrate reporting into existing IT systems; marginal cost per transaction is negligible.	MiFID II transaction reporting; SEC Consolidated Audit Trail; VAT real-time invoicing (Brazil SPED, Italy SDI).
Probabilistic performance standards	Productive firms exhibit lower operational variance and greater redundancy; expected compliance cost converges to zero.	Electric grid reliability (NERC, ENTSO-E); airline on-time performance (US DOT).
Rolling-window compliance	Large firms smooth shocks across time and business units; smaller firms face lumpy, binding constraints.	Bank capital stress tests (CCAR, EBA); emissions averaging (EU ETS compliance windows).
Risk-based supervision	Productive firms invest in systems that drive inspection probability toward zero; legal rules are uniform but expected costs are endogenous.	Banking supervision (Basel, ECB SSM, Fed); OSHA/EPA inspection targeting.
Scope-based administrative burden	Large firms already operate at scope and internalize requirements within existing compliance structures.	GDPR (data mapping, breach response, DPIAs); consumer disclosure rules (mortgages, credit cards).
Endogenously priced insurance mandates	Premiums are market-priced based on firm-specific risk; productive firms face negligible relative cost increases.	Environmental liability insurance (oil, mining); professional malpractice insurance.

In each case, the regulation raises marginal costs industry-wide but the burden becomes asymptotically negligible for the most productive firms. These instruments are uniform in design — they do not rely on exemptions, grandfathering, or discriminatory enforcement — yet they change the shape of the cost distribution and therefore fall within the scope of the mechanism developed in this paper.

References

AARONSON, D. AND E. FRENCH (2007): "Product Market Evidence on the Employment Effects of the Minimum Wage," *Journal of Labor Economics*, 25, 167–200.

ALLEGRETTO, S. AND M. REICH (2018): "Are Local Minimum Wages Absorbed by Price Increases? Estimates from Internet-Based Restaurant Menus," *ILR Review*, 71, 35–63.

BECKER, G. S. (1983): "A Theory of Competition among Pressure Groups for Political Influence," *The Quarterly Journal of Economics*, 98, 371–400.

BERGER, D., K. HERKENHOFF, AND S. MONGEY (2025): "Minimum Wages, Efficiency, and Welfare," *Econometrica*, 93, 265–301.

BOMBARDINI, M. (2008): "Firm Heterogeneity and Lobby Participation," *Journal of International Economics*, 75, 329–348.

BOMBARDINI, M. AND F. TREBBI (2011): "Votes or Money? Theory and Evidence from the U.S. Congress," *Journal of Public Economics*, 95, 587–611.

——— (2020a): "Empirical Models of Lobbying," *Annual Review of Economics*, 12, 391–413.

——— (2020b): "Empirical Models of Lobbying," *Annual Review of Economics*, 12, 391–413.

CRANE, A. AND A. KOCH (2025a): "The Impact of Regulation on Firm Value: Evidence from Political Connections," *The Review of Corporate Finance Studies*, 14, 1058–1082.

——— (2025b): "The Impact of Regulation on Firm Value: Evidence from Political Connections," *The Review of Corporate Finance Studies*, 14, 1058–1082.

DAL BÓ, E. (2006): "Regulatory Capture: A Review," *Oxford Review of Economic Policy*, 22, 203–225.

DICKSON, A. AND I. A. MACKENZIE (2018): "Strategic Trade in Pollution Permits," *Journal of Environmental Economics and Management*, 87, 94–113.

——— (2022): "Permit Markets with Political and Market Distortions," *Environmental and Resource Economics*, 82, 227–255.

DRACA, M., S. MACHIN, AND J. VAN REENEN (2011): "Minimum Wages and Firm Profitability," *American Economic Journal: Applied Economics*, 3, 129–151.

DRECHSEL-GRAU, M. (2023): "Employment and Reallocation Effects of Higher Minimum Wages," Working Paper, LMU Munich.

DU, P. AND S. WANG (2020): "The Effect of Minimum Wage on Firm Markup: Evidence from China," *Economic Modelling*, 86, 241–250.

DUSTMANN, C., A. LINDNER, U. SCHÖNBERG, M. UMKEHRER, AND P. VOM BERGE (2022): "Reallocation Effects of the Minimum Wage," *Quarterly Journal of Economics*, 137, 267–328.

ENGBOM, N. AND C. MOSER (2022): "Earnings Inequality and the Minimum Wage: Evidence from Brazil," *American Economic Review*, 112, 3803–3847.

FAN, Y. AND F. ZHOU (2023a): "Market Power and Political Connections," Working paper, SSRN, sSRN 4350527.

FAN, Y. AND J. ZHOU (2023b): "Market Power and Political Connections," SSRN Working Paper.

GROSSMAN, G. M. AND E. HELPMAN (1994): "Protection for Sale," *American Economic Review*, 84, 833–850.

HAANWINCKEL, D. (2025): "Supply, Demand, Institutions, and Firms: A Theory of Labor Market Sorting and the Wage Distribution," NBER Working Paper 31318.

HAELBIG, M., M. MERTENS, AND S. MÜLLER (2023): "Minimum Wages, Productivity, and Reallocation," IZA Discussion Paper 16160.

HARASZTOSI, P. AND A. LINDNER (2019): "Who Pays for the Minimum Wage?" *American Economic Review*, 109, 2693–2727.

HAU, H., Y. HUANG, AND G. WANG (2020): "Firm Response to Competitive Shocks: Evidence from China's Minimum Wage Policy," *Review of Economic Studies*, 87, 2639–2671.

HAUCAP, J., U. PAULY, AND C. WEY (2001): "The Incentives of Employers' Associations to Raise Rivals' Costs in the Presence of Collective Bargaining," *International Journal of Industrial Organization*, 19, 1589–1618.

HUNEEUS, F. AND I. S. KIM (2024): "The Effects of Firms' Lobbying on Resource Misallocation," Working Paper.

HVIID, M. AND M. OLCZAK (2016): "Raising Rivals' Fixed Costs," *International Journal of the Economics of Business*, 23, 1–26.

KENNARD, A. (2020): "The Enemy of My Enemy: When Firms Support Climate Change Regulation," *International Organization*, 74, 187–221.

LAFFONT, J.-J. AND J. TIROLE (1993): *A Theory of Incentives in Procurement and Regulation*, Cambridge, MA: MIT Press.

LUCA, D. L. AND M. LUCA (2018): "Survival of the Fittest: The Impact of the Minimum Wage on Firm Exit," Harvard Business School NOM Unit Working Paper 17-088.

MACEDONI, L. AND A. WEINBERGER (2024): "Lobbying for Regulations: When Big Business Says Yes," Working Paper.

MARVEL, H. P. (1977): "Factory Regulation: A Reinterpretation of Early English Experience," *Journal of Law and Economics*, 20, 379–402.

——— (1982): "Exclusive Dealing," *Journal of Law and Economics*, 25, 1–25.

MATSUYAMA, K. AND P. USHCHEV (2017): "Three Alternative Classes of Flexible Homothetic Demand Systems," SSRN Working Paper.

——— (2020): "Constant Pass-Through," CEPR Discussion Paper No. 15475 / SSRN Working Paper.

MAYNERIS, F., S. PONCET, AND T. ZHANG (2018): "Improving or Disappearing: Firm-Level Adjustments to Minimum Wages in China," *Journal of Development Economics*, 135, 20–42.

MUEHLEGGER, E. AND R. L. SWEENEY (2022): "Pass-Through of Own and Rival Cost Shocks: Evidence from the U.S. Fracking Boom," *The Review of Economics and Statistics*, 104, 1361–1369.

ORDOVER, J. A., G. SALONER, AND S. C. SALOP (1990): "Equilibrium Vertical Foreclosure," *American Economic Review*, 80, 127–142.

PELTZMAN, S. (1976): "Toward a More General Theory of Regulation," *Journal of Law and Economics*, 19, 211–240.

RAO, N. AND M. RISCH (2026): "Who's Afraid of the Minimum Wage?" *Quarterly Journal of Economics*, 141, 373–423.

RILEY, R. AND C. ROSAZZA BONDIBENE (2017): "Raising the Standard: Minimum Wages and Firm Productivity," *Labour Economics*, 44, 27–50.

SALOP, S. C. AND D. T. SCHEFFMAN (1983): "Raising Rivals' Costs," *American Economic Review*, 73, 267–271.

——— (1987): "Cost-Raising Strategies," *Journal of Industrial Economics*, 36, 19–34.

STIGLER, G. J. (1971): "The Theory of Economic Regulation," *The Bell Journal of Economics and Management Science*, 2, 3–21.

SWEENEY, R. L. (2022): "Environmental Regulation, Imperfect Competition and Market Spillovers: The Impact of the 1990 Clean Air Act Amendments on the US Oil Refining Industry," Working Paper WP 23-19, Resources for the Future.

TULLOCK, G. (1967): "The Welfare Costs of Tariffs, Monopolies, and Theft," *Western Economic Journal*, 5, 224–232.

WILLIAMSON, O. E. (1968): "Wage Rates as a Barrier to Entry: The Pennington Case in Perspective," *Quarterly Journal of Economics*, 82, 85–116.